

A Berger-Five-Sphere Counterexample to Sharp Spectral and Contraction Comparison

Candidate full counterexample packet
arXiv:1508.00606v4, Conjectures 3 and 4

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Status. Candidate full counterexample to both Conjectures 3 and 4 of Milman [1], subject to expert review. The final spectral certificate is exact and occurs on a standard homogeneous Berger metric on S^5 .

1 Source questions

Let g_{can}^ρ be the round metric on S^n rescaled so that $\text{Ric}_{g_{\text{can}}^\rho} = \rho g_{\text{can}}^\rho$. Milman's Conjecture 3 asks whether every metric g on S^n satisfying $\text{Ric}_g \geq \rho g$ obeys

$$\lambda_j(S^n, g) \geq \lambda_j(S^n, g_{\text{can}}^\rho) \quad (j \geq 1), \quad (1)$$

where the source indexes the spectrum with $\lambda_1 = 0$. Conjecture 4 asks more strongly for a metric contraction

$$T : (S^n, g_{\text{can}}^\rho, \text{vol}_{g_{\text{can}}^\rho}) \longrightarrow (S^n, g, \text{vol}_g)$$

that pushes the first volume measure to a finite constant times the second [1, pp. 8–9]. Such a map implies (1) by the source's Contraction Principle.

ric), the natural model space is simply the canonical n -sphere S^n with its metric g_{can}^ρ rescaled to have $\text{Ric}_{g_{\text{can}}^\rho} = \rho g_{\text{can}}^\rho$. For similar topological reasons as in the previous subsection (see also the ensuing discussion), we restrict to the case when (M^n, g) is diffeomorphic to a sphere.

Conjecture 3 (Spectral Comparison Conjecture For Positively Curved (S^n, g, vol_g)).
For any (S^n, g, vol_g) satisfying $\text{Ric}_g \geq \rho g$ with $\rho > 0$, we have:

$$\forall k \geq 1 \quad \lambda_k(S^n, g, \text{vol}_g) \geq \lambda_k(S^n, g_{\text{can}}^\rho, \text{vol}_{g_{\text{can}}^\rho}).$$

Conjecture 3 is consistent with:

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Figure 1: Conjecture 3 in the source PDF, page 8.

2 The Berger metric and proof intuition

View the unit round sphere $S^5 \subset \mathbb{C}^3$ as the total space of the complex Hopf fibration $S^1 \rightarrow S^5 \rightarrow \mathbb{C}P^2$. Its tangent bundle splits orthogonally as $TS^5 = \mathcal{H} \oplus \mathcal{V}$, where $\mathcal{V}$ is generated by the unit Hopf field $\xi(z) = iz$. If g_0 is the unit round metric, put

$$g_a = g_0|_{\mathcal{H}} + a g_0|_{\mathcal{V}}, \quad a = \frac{5}{4}. \quad (2)$$

Thus Hopf-fiber lengths are multiplied by $\sqrt{a}$.

Conjecture 3 would follow immediately from the Contraction Principle and the following previously unpublished conjecture of ours [65]:

Conjecture 4 (Contraction Conjecture for Positively Curved (S^n, g, vol_g)). *For any (S^n, g, vol_g) satisfying $\text{Ric}_g \geq \rho g$ with $\rho > 0$, there exists a map:*

$$T : (S^n, g_{\text{can}}^\rho, \text{vol}_{g_{\text{can}}^\rho}) \rightarrow (S^n, g, \text{vol}_g),$$

pushing forward $\text{vol}_{g_{\text{can}}^\rho}$ onto vol_g up to a finite constant and contracting the corresponding metrics.

Note that a connected complete Riemannian manifold (M, g) with $\text{Ric}_g \geq \rho g$, $\rho > 0$, is necessarily compact and has finite volume. The reader should note the apparent analogy between the latter conjecture and Caffarelli's Contraction Theorem, in view of the definition of the generalized Ricci tensor (1.1). Conjecture 4 is consistent with:

- The Bonnet–Meyers bound on the diameter of such manifolds [36]:

$$\text{diam}(M) \leq \text{diam}(S^n, g_{\text{can}}^\rho) = \frac{\pi}{\sqrt{\rho/(n-1)}}.$$

- The Bishop–Croke volume estimate [27].

Figure 2: The implication from Conjecture 4 to Conjecture 3 and the statement of Conjecture 4, source PDF page 9.

Proof intuition. Stretching the Hopf fibers has two competing effects. It decreases the eigenvalues of functions with high circle weight, but it also decreases the horizontal Ricci curvature and hence enlarges the round comparison sphere. At low degrees the comparison rescaling wins. At degree seven, however, the purely holomorphic modes have enough Hopf weight to move below the corresponding round level. The degree-six and degree-seven clusters remain separated, so a finite multiplicity count turns this crossing into an exact violation of the ordered spectrum at index 715.

3 Geometric and spectral formulas

We record the two standard canonical-variation calculations in the precise normalization used here. They may also be obtained from the Gray–O'Neill formulas and the classical Berger-sphere spectral decomposition [2, 3].

Lemma 1 (Ricci tensor). *For S^{2m+1} with $g_a = g_{\mathcal{H}} + ag_{\mathcal{V}}$, the Ricci eigenvalues relative to g_a are*

$$r_{\mathcal{H}} = 2m + 2 - 2a, \quad r_{\mathcal{V}} = 2ma.$$

For (2), they are respectively 7/2 and 5.

Proof. For a g_a -unit horizontal vector X and a g_a -unit vertical vector U , the canonical-variation sectional curvatures are

$$K_a(X, U) = a, \quad K_a(X, JX) = 4 - 3a,$$

and $K_a(X, Y) = 1$ for horizontal $Y \perp \{X, JX\}$. Summing over an orthonormal frame gives

$$\text{Ric}_a(X, X) = (4 - 3a) + (2m - 2) + a = 2m + 2 - 2a, \quad \text{Ric}_a(U, U) = 2ma.$$

Putting $m = 2$ and $a = 5/4$ gives 7/2 and 5. □

Lemma 2 (Function spectrum). *Let $\mathcal{H}_{p,q}$ be the restrictions to S^{2m+1} of harmonic polynomials on $\mathbb{C}^{m+1}$ that are homogeneous of degree p in z and degree q in $\bar{z}$. On $\mathcal{H}_{p,q}$, the positive Laplacian of g_a has eigenvalue*

$$\Lambda_a(p, q) = k(k + 2m) + (a^{-1} - 1)(p - q)^2, \quad k = p + q. \quad (3)$$

The direct sum over $p + q = k$ is the full round degree- k spherical-harmonic space.

Proof. The positive round Laplacian has eigenvalue $k(k + 2m)$ on degree- k spherical harmonics. The Hopf generator satisfies $\xi f = i(p - q)f$ on $\mathcal{H}_{p,q}$. Scaling the vertical squared length by a changes the positive function Laplacian by $(a^{-1} - 1)(-\xi^2)$, which gives (3). The usual complex bihomogeneous decomposition of harmonic polynomials gives the last assertion. □

4 Exact counterexample

Theorem 1. *The Berger metric $g_{5/4}$ on S^5 satisfies $\text{Ric}_{g_{5/4}} \geq (7/2)g_{5/4}$, but*

$$\lambda_{715}(S^5, g_{5/4}) = \frac{336}{5} < \frac{539}{8} = \lambda_{715}(S^5, g_{\text{can}}^{7/2}).$$

The deficit is $7/40$. Consequently Conjectures 3 and 4 of [1] are both false.

Proof. The Ricci assertion is Lemma 1. Since the unit round S^5 has $\text{Ric} = 4g_0$, its $(7/2)$ -Ricci comparison metric is $(8/7)g_0$. Its positive Laplacian therefore has degree- k eigenvalue

$$\Lambda_{\text{cmp}}(k) = \frac{7}{8}k(k+4). \quad (4)$$

For $a = 5/4$, Lemma 2 becomes

$$\Lambda_{5/4}(p, q) = k(k+4) - \frac{1}{5}(p-q)^2. \quad (5)$$

The largest eigenvalue among all modes of degrees $0 \leq k \leq 6$ is at most $6(6+4) = 60$. For fixed degree k , the smallest value in (5) occurs at $|p-q| = k$ and equals

$$F(k) = k(k+4) - \frac{1}{5}k^2 = \frac{4}{5}k^2 + 4k.$$

This is strictly increasing for $k \geq 0$, and $F(7) = 336/5 > 60$. Hence every mode of degree at most six precedes every mode of degree at least seven in the ordered $g_{5/4}$ spectrum, and the first degree-seven value is $336/5$. It is attained, for instance, on nonzero holomorphic homogeneous polynomials of degree seven, which are harmonic.

The dimension of the degree- k spherical-harmonic space on S^5 is

$$d_k = \binom{k+5}{5} - \binom{k+3}{5},$$

with the second term interpreted as zero for $k < 2$. For $k = 0, \dots, 6$ this gives

$$(d_0, \dots, d_6) = (1, 6, 20, 50, 105, 196, 336), \quad \sum_{k=0}^6 d_k = 714.$$

Remembering that the source convention starts with $\lambda_1 = 0$, we obtain

$$\lambda_{715}(S^5, g_{5/4}) = F(7) = \frac{336}{5}.$$

The comparison sphere has the same degree multiplicities and strictly increasing levels (4), so its 715th eigenvalue is its first degree-seven value:

$$\lambda_{715}(S^5, g_{\text{can}}^{7/2}) = \frac{7}{8}7(7+4) = \frac{539}{8}.$$

Finally,

$$\frac{539}{8} - \frac{336}{5} = \frac{7}{40} > 0,$$

which disproves Conjecture 3.

For completeness, suppose a map from the comparison sphere to $(S^5, g_{5/4})$ as in Conjecture 4 existed. Pullback by this map is injective on L^2 because its pushforward is a positive constant times target volume. The metric contraction and the almost-everywhere chain rule give

$$\int |\nabla(f \circ T)|^2 d\text{vol}_{g_{\text{can}}^{7/2}} \leq c \int |\nabla f|^2 d\text{vol}_{g_{5/4}}, \quad \int |f \circ T|^2 d\text{vol}_{g_{\text{can}}^{7/2}} = c \int |f|^2 d\text{vol}_{g_{5/4}}$$

for the same $c > 0$. The min-max principle would imply every comparison eigenvalue is at most the corresponding target eigenvalue, contradicting the 715th-eigenvalue calculation. Thus Conjecture 4 is false as well. $\square$

5 Verification, scope, and novelty

The script `code/verify_spectrum.py` uses exact rational arithmetic. It checks the two Ricci values, the 7/8 comparison scale, the bihomogeneous multiplicities, the 714-mode count, degree separation, both 715th eigenvalues, and the exact 7/40 deficit. Its final output is:

```
Ricci eigenvalues (horizontal, vertical) = 7/2 5
modes of degrees 0 through 6 = 714
lambda_715(Berger S^5) = 336/5
lambda_715(round comparison S^5) = 539/8
strict deficit = 7/40
all exact checks passed
```

The construction refutes the universal conjectures but does not classify the dimensions or metric families where comparison survives. In particular, Lin, Wang, and Xu recently proved Conjecture 3 for all positively curved two-spheres [4], and Serres proved a near-round two-sphere case of the contraction statement [5]; this example is five-dimensional and does not conflict with either result. It also does not contradict the source’s heat trace comparison or its asymptotic Weyl comparison: coordinatewise spectral ordering is strictly stronger.

A bounded novelty search through 11 August 2026 covered the four run indexes; the exact source id, title, and conjecture phrases; combinations of Milman’s name with “Berger sphere”, “spectral comparison”, and “counterexample”; citation-oriented searches; the 2018 journal page; and the two recent two-sphere papers just cited. It found no prior counterexample or this calculation. This is not a claim of priority.

Human-review recommendation. Very high priority. Verify Lemmas 1 and 2 in the stated fiber-scaling convention. Once those standard formulas are fixed, the remaining proof is a short exact multiplicity count.

References

- [1] E. Milman, *Spectral estimates, contractions and hypercontractivity*, J. Spectr. Theory 8 (2018), 669–714; [arXiv:1508.00606v4](#).
- [2] A. L. Besse, *Einstein Manifolds*, Springer, 1987, discussion of canonical variations and Berger spheres.
- [3] S. Tanno, *The first eigenvalue of the Laplacian on spheres*, Tohoku Math. J. 31 (1979), 179–185.
- [4] S. Lin, H. Wang, and G. Xu, *Eigenvalues on spheres*, [arXiv:2607.11544](#) (2026).
- [5] J. Serres, *Contractive transport maps from S^2 to nearly spherical surfaces with positive Ricci curvature*, [arXiv:2508.13688](#) (2025).